DECISION SUPPORT SYSTEM COURSE JOBSHEET

WEEK : 7
MATERIAL : Evaluation Based On Distance From Average Solution (EDAS)
PURPOSE : Students are able to apply the EDAS method to decision-making problems

METHOD DISCUSSION

This method is especially useful when conflicting criteria must be considered. As has been discovered and claimed by the authors of the method, the EDAS method is stable when various weight criteria are used, and is consistent with other methods. In addition, its simplicity and faster calculations are advantages of those proposed by the authors of this method, especially since these advantages do not affect the accuracy of calculations.

STAGES OF THE EDAS METHOD

The *Evaluation method based on Distance from Average Solution* / EDAS was introduced by *KeshavarazvvGhorabaee, et all* in 2015. In solving problems and rankings, the EDAS method has several steps, namely:

1. Formation of the result matrix (X)

   In the results matrix (X), the row shows Alternatives and the column shows the criteria. The decision matrix shows the performance of each alternative against various criteria.

   $$X = \begin{bmatrix} x_{01} & \cdots & x_{0j} & \cdots & x_{0n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i1} & \cdots & x_{ij} & \cdots & x_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{m1} & \cdots & x_{m2} & \cdots & x_{mn} \end{bmatrix} (i = 0, 1, 2, \ldots, m; j = 1, 2, \ldots, n)$$

   xij shows the performance value of the first alternative in the Jth criterion, *M* is the number of alternatives while *N* is the number of criteria.

2. Determining the average solution (AV)

   The average solution (AV) corresponds to the criteria, determined by the equation:

   $$AV = [AV_j]_{1\times m}$$

   where the value of AVj can be calculated with the equation:

   $$AV_j = \frac{\sum_{i=1}^{n} X_{ij}}{n}$$

3. Determining positive/negative distance from average (PDA/NDA)

Calculate the positive distance from the mean matrix (PDA) and the negative distance from the mean matrix (NDA) according to the type of criteria (*benefit* and *cost*) using the equation:

$$PDA = [PDA_{ij}]_{n\times m}$$

$$NDA = [NDA_{ij}]_{n\times m}$$

For the j criterion which is a *Benefit type* criterion, it applies:

$$PDA_{ij} = \frac{max(0,(X_{ij}-AV_j))}{AV_j}$$

$$NDA_{ij} = \frac{max(0,(AV_j-X_{ij}))}{AV_j}$$

As for the j criterion which is a Cost type criterion , it applies:

$$PDA_{ij} = \frac{max(0,(AV_j-X_{ij}))}{AV_j}$$

$$NDA_{ij} = \frac{max(0,(X_{ij}-AV_j))}{AV_j}$$

4. Determining the weighted amount of PDA/NDA (SP/SN)

   Determine the weighted amount of PDA and NDA for all alternatives with the following equation:

$$SP_i = \sum_{j=1}^{m} w_j \times PDA_{ij}$$

$$SN_i = \sum_{j=1}^{m} w_j \times NDA_{ij}$$

   The SPi and SNi values, respectively, are the weighted sum values of PDA and NDA for each ith alternative.

5. Normalization of SP/SN (NSP/NSN) values

   The next stage is to calculate the normalization value of the SP and SN for all alternatives.

$$NSP_i = \frac{SP_i}{max_i(SP_i)}$$

$$NSN_i = 1 - \frac{SN_i}{max_i(SN_i)}$$

6. Calculate the value of the assessment score (US)

   After the normalization values of NSP and NSN are obtained, the Apraisal Score - *Assessment Value* - AS is calculated as follows:

$$AS_i = \frac{1}{2}(NSP_i + NSN_i);\ \text{dengan}\ 0 \le AS_i \le 1$$

7. Ranking

The last stage is the ranking of the US assessment score from the highest to the lowest. The alternative with the highest value indicates the best alternative.

EXAMPLE CASES & CALCULATIONS

In recent years, there has been a high increase in attention to sustainable transportation. With congestion issues at the top of the list of important issues in medium and large cities, the diverse means of mass transportation, and especially commuter parking systems, have become one of the most discussed sustainable transportation alternatives.

Commuter parking facilities offer the possibility to access the city center using public transportation. It can be buses, rapid transit buses, trains, or other integrated modes of transportation. The commuter parking system can be explained in simple terms: people use private vehicles to drive from their residence to a commuter parking facility, park there, and switch to public transportation to reach their destination.

The use of *the Evaluation based on Distance from Average Solution* (EDAS) method is expected to help policymakers to determine the best commuter parking location, based on predetermined criteria and existing location alternatives. There are 7 alternative locations to **be** chosen, namely *Lot 51, Plot 61, Plot 77, Area 51, Lot 61, Plot 57, and Complex 51.*

The details of the assessment weight are as shown in table 1 below:

Table 1. Criteria and weights for choosing a commuter parking location

Kode	Kriteria	Tipe[1]	Bobot[2]
C_1	Luas Lahan	benefit	0.133
C_2	Jarak ke Pusat Kota	cost	0.053
C_3	Sistem Informasi Pendukung	benefit	0.080
C_4	Keunggulan Transportasi Umum dibanding Angkutan Pribadi	benefit	0.067
C_5	Promosi oleh Institusi Umum	benefit	0.027
C_6	Frekuensi Angkutan Umum di lokasi	benefit	0.093
C_7	Harga Tempat Parkir	cost	0.187
C_8	Trafik Angkutan Umum di Lokasi	cost	0.107
C_9	Lokasi Parkir Gratis di Pusat Kota	cost	0.040
C_{10}	Total Biaya Parkir dan Angkutan Umum	cost	0.160

The initial data that will be taken into account with this EDAS method are as listed in the following table 2:

Table 2. Sample data

Alternatif		Kriteria									
Kode	Nama	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
A_1	Lot 51	4	3	7	6	8	6	4	4	5	4
A_2	Petak 61	2	4	4	8	5	6	5	2	3	3
A_3	Petak 77	4	3	7	9	7	4	5	3	4	3
A_4	Area 51	4	5	8	8	8	3	5	4	3	3
A_5	Lot 61	4	4	4	9	6	3	3	3	5	3
A_6	Petak 57	2	3	6	8	7	3	3	4	5	3
A_7	Komplek 51	4	5	4	9	5	6	4	3	3	4

The following will be described the calculation using the EDAS method step by step:

1. Matrix of results (X)

The first step is to create a decision matrix (X) from the initial data that exists. From the data in Table 2, the following decision matrix can be made:

$$X = \begin{bmatrix} 4 & 3 & 7 & 6 & 8 & 6 & 4 & 4 & 5 & 4 \\ 2 & 4 & 4 & 8 & 5 & 6 & 5 & 2 & 3 & 3 \\ 4 & 3 & 7 & 9 & 7 & 4 & 5 & 3 & 4 & 3 \\ 4 & 5 & 8 & 8 & 8 & 3 & 5 & 4 & 3 & 3 \\ 4 & 4 & 4 & 9 & 6 & 3 & 3 & 3 & 5 & 3 \\ 2 & 3 & 6 & 8 & 7 & 3 & 3 & 4 & 5 & 3 \\ 4 & 5 & 4 & 9 & 5 & 6 & 4 & 3 & 3 & 4 \end{bmatrix}$$

In the decision matrix (X), the data in the ith row shows the data from the ith Alternative; while the data in the jth column shows the jth criterion. For example, for data x2.6 shows data for alternative 2, namely **Plot 61** for criterion 6 (*Frequency of Public Transportation at the location*) with a value of 6.

2. Average Solution Value (AV)

The Average Solution Value (AV) of each criterion is calculated from the average value of all alternatives on a given criterion. For example, for the 6th criterion (*Public Transportation Frequency at the location*), the average solution value (AV6) can be calculated as follows.

$$\begin{aligned} AV_6 &= \frac{\sum_{i=1}^{7} x_{i,6}}{7} \\ &= \frac{x_{1,6} + x_{2,6} + x_{3,6} + x_{4,6} + x_{5,6} + x_{6,6} + x_{7,6}}{7} \\ &= \frac{6+6+4+3+3+3+6}{7} \\ &= \frac{31}{7} \\ &= 4.429 \end{aligned}$$

With the same calculation, the average AV solution value is calculated for other criteria so that the AV average solution value matrix is obtained as follows:

$$AV = [3.429 \quad 3.857 \quad 5.714 \quad 8.143 \quad 6.571 \quad 4.429 \quad 4.143 \quad 3.286 \quad 4.000 \quad 3.286]$$

3. Calculating Positive/Negative Distance from Mean (PDA/NDA)

After obtaining the average solution value (AV), the next is to calculate the value of the Positive/Negative distance from the average (PDA/NDA). For the benefit type criteria , based on the equation in step number 3, the PDA/NDA value can be searched. For example, for the 1st criterion (*Land Area*) in the 2nd alternative (**Plot 61), the** values of PDA2.1 and NDA2.1 can be calculated as follows:

$$\begin{aligned}
PDA_{2,1} &= \max(0, \frac{(X_{2,1} - AV_1)}{AV_1}) \\
&= \max(0, \frac{(2.000 - 3.429)}{3.429}) \\
&= \max(0, \frac{-1.429}{3.429}) \\
&= \max(0, \ -0.417) \\
&= 0 \\
NDA_{2,1} &= \max(0, \frac{(AV_1 - X_{2,1})}{AV_1}) \\
&= \max(0, \frac{(3.429 - 2.000)}{3.429}) \\
&= \max(0, \frac{1.429}{3.429}) \\
&= \max(0, \ 0.417) \\
&= 0.417
\end{aligned}$$

As for the cost type criteria *, the* PDA/NDA value can be searched based on the equation in step number 3 at the bottom. For example, for the 8th criterion (Public *Transportation Traffic at the Location*) in the 2nd alternative (**Plot 61**) the values of PDA2.8 and NDA2.8 can be calculated as follows:

$$
\begin{aligned}
PDA_{2,8} &= \max(0,\ \frac{(AV_8 - X_{2,8})}{AV_8}) \\
&= \max(0,\ \frac{(3.286 - 2.000)}{3.286}) \\
&= \max(0,\ \frac{1.286}{3.286}) \\
&= \max(0,\ 0.391) \\
&= 0.391 \\
NDA_{2,8} &= \max(0,\ \frac{(X_{2,8} - AV_8)}{AV_8}) \\
&= \max(0,\ \frac{(2.000 - 3.286)}{3.286}) \\
&= \max(0,\ \frac{-1.286}{3.286}) \\
&= \max(0,\ -0.391) \\
&= 0
\end{aligned}
$$

With the same calculation for all data from the decision matrix (X), positive distance data from the mean solution (PDA) is obtained as in the following Table 3:

Table 3. Positive distance from average solution (PDA)

No.	Alternatif		PDA									
	Kode	Nama	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
1	A_1	Lot 51	0.167	0.222	0.225	0.000	0.217	0.355	0.034	0.000	0.000	0.000
2	A_2	Petak 61	0.000	0.000	0.000	0.000	0.000	0.355	0.000	0.391	0.250	0.087
3	A_3	Petak 77	0.167	0.222	0.225	0.105	0.065	0.000	0.000	0.087	0.000	0.087
4	A_4	Area 51	0.167	0.000	0.400	0.000	0.217	0.000	0.000	0.000	0.250	0.087
5	A_5	Lot 61	0.167	0.000	0.000	0.105	0.000	0.000	0.276	0.087	0.000	0.087
6	A_6	Petak 57	0.000	0.222	0.050	0.000	0.065	0.000	0.276	0.000	0.000	0.087
7	A_7	Komplek 51	0.167	0.000	0.000	0.105	0.000	0.355	0.034	0.087	0.250	0.000

As for the negative distance data from the average solution (NDA) as shown in Table 4 below:

Table 4. Negative distance from mean solution (PDA)

No.	Alternatif		NDA									
	Kode	Nama	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
1	A_1	Lot 51	0.000	0.000	0.000	0.263	0.000	0.000	0.000	0.217	0.250	0.217
2	A_2	Petak 61	0.417	0.037	0.300	0.018	0.239	0.000	0.207	0.000	0.000	0.000
3	A_3	Petak 77	0.000	0.000	0.000	0.000	0.000	0.097	0.207	0.000	0.000	0.000
4	A_4	Area 51	0.000	0.296	0.000	0.018	0.000	0.323	0.207	0.217	0.000	0.000
5	A_5	Lot 61	0.000	0.037	0.300	0.000	0.087	0.323	0.000	0.000	0.250	0.000
6	A_6	Petak 57	0.417	0.000	0.000	0.018	0.000	0.323	0.000	0.217	0.250	0.000
7	A_7	Komplek 51	0.000	0.296	0.300	0.000	0.239	0.000	0.000	0.000	0.000	0.217

4. Calculating the weighted amount of PDA/NDA (SP/SN)

The SP/SN value is the sum of the PDA/NDA value multiplied by the weight (w) according to the criteria for each alternative. For example, SP2 is the sum of the multiplication of the value of PDA2.j with the weight of the criterion of j (wj). In accordance with the equation in step number 4, the SP2 value is obtained as follows:

$$
\begin{aligned}
SP_2 &= \sum_{j=1}^{10} w_j \times PDA_{2,j} \\
&= w_1 * PDA_{2,1} + w_2 * PDA_{2,2} + w_3 * PDA_{2,3} + w_4 * PDA_{2,4} + w_5 * PDA_{2,5} + w_6 * PDA_{2,6} + w_7 * PDA_{2,7} + w_8 * PDA_{2,8} + w_9 * PDA_{2,9} + w_{10} * PDA_{2,10} \\
&= 0.133 * 0.000 + 0.053 * 0.000 + 0.080 * 0.000 + 0.067 * 0.000 + 0.027 * 0.000 + 0.093 * 0.355 + 0.187 * 0.000 + 0.107 * 0.391 + 0.040 * 0.250 + 0.160 * 0.087 \\
&= 0.000 + 0.000 + 0.000 + 0.000 + 0.000 + 0.033 + 0.000 + 0.042 + 0.010 + 0.014 \\
&= 0.098770453482936
\end{aligned}
$$

While the SN2 value is the sum of the multiplication of the value of NDA2.j with the weight of the j criterion (wj), according to the equation EDA-11 can be calculated as follows:

$$
\begin{aligned}
SN_2 &= \sum_{j=1}^{10} w_j \times NDA_{2,j} \\
&= w_1 * NDA_{2,1} + w_2 * NDA_{2,2} + w_3 * NDA_{2,3} + w_4 * NDA_{2,4} + w_5 * NDA_{2,5} + w_6 * NDA_{2,6} + w_7 * NDA_{2,7} + w_8 * NDA_{2,8} + w_9 * NDA_{2,9} + w_{10} * NDA_{2,10} \\
&= 0.133 * 0.000 + 0.053 * 0.000 + 0.080 * 0.000 + 0.067 * 0.000 + 0.027 * 0.000 + 0.093 * 0.355 + 0.187 * 0.000 + 0.107 * 0.391 + 0.040 * 0.250 + 0.160 * 0.087 \\
&= 0.056 + 0.002 + 0.024 + 0.001 + 0.006 + 0.000 + 0.039 + 0.000 + 0.000 + 0.000 \\
&= 0.12769795609018
\end{aligned}
$$

After the same calculation is made for the other alternatives, the SP/SN value is obtained as in the following Table 5:

Table 5. Weighted amount of PDA/NDA (SP/SN)

No.	Alternatif		SP	SN
	Kode	Nama		
1	A_1	Lot 51	0.09743	0.08551
2	A_2	Petak 61	0.09877	0.12770
3	A_3	Petak 77	0.08402	0.04765
4	A_4	Area 51	0.08393	0.10889
5	A_5	Lot 61	0.10392	0.06840
6	A_6	Petak 57	0.08300	0.12002
7	A_7	Komplek 51	0.08807	0.08096

5. Calculating the normalization value of SP/SN (NSP/NSN)

The value of the weighted sum of SP/SN of each alternative is then normalized by dividing it by the maximum value of its SP/SN value. For example, for the NSP2 value -- the NSP value of the 2nd alternative (**Plot 61**) -- is calculated based on the equation the SP2 value divided by the max(SP) value, as follows:

$$
\begin{aligned}
NSP_2 &= \frac{SP_2}{\max(SP)} \\
&= \frac{0.09877}{0.10392} \\
&= 0.9504248358151
\end{aligned}
$$

$$
\begin{aligned}
NSN_2 &= 1 - \frac{SN_2}{\max(SN)} \\
&= 1 - \frac{0.12770}{0.12770} \\
&= 0
\end{aligned}
$$

For the other alternatives, the NSP/NSN value can be calculated in the same way, and the result is as follows:

Table 6. Normalized values (NSP/NSN)

No.	Alternatif		NSP	NSN
	Kode	Nama		
1	A_1	Lot 51	0.93749	0.33033
2	A_2	Petak 61	0.95042	0.00000
3	A_3	Petak 77	0.80848	0.62683
4	A_4	Area 51	0.80764	0.14730
5	A_5	Lot 61	1.00000	0.46435
6	A_6	Petak 57	0.79866	0.06012
7	A_7	Komplek 51	0.84746	0.36599

6. Calculate the value of the assessment score (US)

The assessment score or *Apraisal Score* (AS) is based on the equation in step number 6, for example the assessment score of the 2nd alternative (**Plot 61**) can be calculated as follows:

$$
\begin{aligned}
AS_{10} &= \frac{1}{2}(NSP_{10} + NSN_{10}) \\
&= \frac{1}{2}(0.24049 + 0.00000) \\
&= \frac{1}{2} * 0.24049 \\
&= 0.12024624741543
\end{aligned}
$$

By calculating all the US values of all existing alternatives, the results are obtained as seen in Table 7 below:

Table 7. Assessment Score (US)

No	Alternatif		AS
	Kode	Nama	
1	A_1	Lot 51	0.63391242547646
2	A_2	Petak 61	0.47521241790755
3	A_3	Petak 77	0.71765515591787
4	A_4	Area 51	0.47746972713864
5	A_5	Lot 61	0.73217395877856
6	A_6	Petak 57	0.42938678262389
7	A_7	Komplek 51	0.60672501623902

7. Ranking

The *Apraisal Score* (AS) score obtained from the results of the previous calculation is then sorted from the largest to the lowest as seen in Table 8 as follows:

Table 8. Ranking

No	Alternatif		AS	Ranking
	Kode	Nama		
1	A_7	Petak 11	0.87346358089148	1
2	A_9	Komplek 11	0.73624560173294	2
3	A_3	Blok 61	0.68253700136596	3
4	A_8	Petak 47	0.5472835699579	4
5	A_2	Area 61	0.51002113033885	5
6	A_4	Komplek 61	0.47365284774861	6
7	A_5	Lot 61	0.47221680268812	7
8	A_1	Petak 13	0.45189274954511	8
9	A_{11}	Petak 29	0.39392334439917	9
10	A_{12}	Blok 53	0.27703163798437	10
11	A_6	Area 47	0.22054191086066	11
12	A_{10}	Lot 57	0.12024624741543	12

From the results of the ranking of the assessment score (AS), it was obtained that the 7th location (**Plot 11**) with an assessment score of 0.87346358089148 was selected as the best location as a commuter parking location, based on the criteria and weights that have been determined.

PRACTICUM ASSIGNMENTS

CASE STUDY: HOME SELECTION

A software developer created a Decision Support System to choose a house using the EDAS method. There are 20 housing that are used as alternatives, while for criteria there are 8 criteria used in decision-making, namely:

Yes	Criterion	Weight
1	Land Area (K1)	0,2904
2	Price (K2)	0,1708
3	Type (K3)	0,2133
4	Water Source (K4)	0,0441
5	Bedroom (K5)	0,1399
6	Bathroom (K6)	0,0293
7	Security Guard Post (K7)	0,0498
8	Location (K8)	0,0624

The scores of each alternative in each criterion are shown in the following decision matrix:

Criterion / Alternative	K1	K2	K3	K4	K5	K6	K7	K8
A1	7	7	9	7	1	9	1	6
A2	4	3	5	3	3	10	7	4
A3	5	5	7	4	9	2	6	1
A4	9	5	10	9	7	8	6	7
A5	10	1	9	2	10	6	10	2
A6	6	7	2	7	10	2	8	7
A7	10	9	10	2	6	7	10	4
A8	7	6	6	4	7	2	3	9
A9	4	9	9	1	6	1	6	1
A10	9	8	2	4	5	3	2	5
A11	4	4	2	5	4	10	6	5
A12	10	6	9	9	10	10	2	6
A13	6	8	8	5	4	9	2	3
A14	8	6	3	3	6	7	8	7
A15	4	9	3	5	10	5	10	6
A16	6	2	7	6	5	6	5	9
A17	8	4	1	8	3	10	7	4
A18	4	1	8	5	10	5	2	6
A19	9	8	9	5	1	3	10	6
A20	3	1	10	4	7	5	10	8

PRACTICUM PROCEDURE

1. Determine the type of criteria out of all the criteria!
2. Make the final score calculation of each alternative using Excel so that it becomes semi-automated and determine the best alternative for the case!
3. **Bonus!** Create a program code for the case study. You can use a programming language according to your skill set!